

ENVIREX / PRITHVEX Executive Report

Environmental Risk Intelligence for Resilient Supply Chains · Team Envirex

Executive Summary

ENVIREX is an environmental risk-intelligence platform for resilient supply chains, built around PRITHVEX, its core calibrated AI engine. The platform combines environmental risk intelligence, calibrated forecasting, explainable machine learning, and autonomous recommendations to take a supply chain organisation from disruption awareness to a quantified mitigation action in roughly three seconds. The remainder of this report summarises the problem, the solution, the measured impact, and the roadmap ahead.

The Problem

Metric	What it shows
\$9.6B	Estimated trade value disrupted in six days during the 2021 Suez Canal blockage
73%	Share of supply chain leaders who report lacking real-time risk data
3-5 Days	Typical organisational lag between a disruption and a coordinated response
3 Seconds	PRITHVEX's measured response time from data ingestion to recommendation

The underlying issue is not a shortage of data; it is the absence of a decision-support layer that turns raw uncertainty into something a planner can act on with confidence.

The Solution

PRITHVEX runs a six-stage pipeline — data ingestion, an ML ensemble, conformal prediction, SHAP attribution, network intelligence, and an action playbook — entirely with zero cloud dependency and full reproducibility. Every stage of the pipeline is exposed through the ENVIREX Command Center, a live dashboard covering a global risk heatmap, a real-time alerts feed, SHAP-based risk attribution, confidence-banded demand forecasts, ranked recommended actions, and an environmental-impact tracker.

Live Demo Scenario: Typhoon Haikui

In the platform's reference demo, a simulated Typhoon Haikui shock is injected into the system. Within three seconds, twelve SKUs flip to high-risk status. The SHAP attribution layer identifies Vietnam port congestion as the top contributing driver, and PRITHVEX auto-generates a recommendation for an emergency stock build across the four most exposed SKUs, with the cost of inaction quantified in dollars rather than left as an abstract risk label.

Measured & Projected Impact

Outcome	Magnitude
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Emergency procurement cost reduction	30-40%
Stockout frequency reduction	~25%
Uncertainty-to-action pipeline time	3 seconds (vs. ~3 days previously)
Carbon footprint reduction	~34% vs. prior month
Disruption response time improvement	~40% faster
Model accuracy improvement from calibration	~30%

Technology Stack

- AI / ML core — XGBoost, LightGBM, scikit-learn, pandas, NumPy, LSTM, Transformers.
- Uncertainty engine — MAPIE conformal prediction with KL-divergence drift monitoring and adaptive recalibration.
- Explainability — SHAP feature attribution with plain-language risk alerts.
- Dashboard — Streamlit, Plotly, NetworkX, and a Monte Carlo simulation engine.
- Infrastructure — zero cloud dependency, fully reproducible, CSV and API data ingest.

Sustainability Alignment

ENVIREX maps directly onto five UN Sustainable Development Goals: SDG 9 (resilient infrastructure, by preventing cascading industrial failures), SDG 12 (responsible consumption, via calibrated order buffers), SDG 13 (climate action, by treating weather and geo-risk indices as first-class risk drivers), SDG 17 (partnerships, through shared multi-tier supplier intelligence), and SDG 8 (decent work, by replacing reactive firefighting with strategic planning).

Roadmap

Initiative	Timeline
Satellite integration (ESA Copernicus) for deforestation & flood monitoring	Round 3
Live AIS port & vessel tracking	Round 3
Mobile alert system for procurement managers	Round 3
Carbon ledger for Scope 3 emissions	Q3 2026
Federated learning across supplier networks	Q3 2026
Regulatory compliance AI (CSRD, SEC Climate, TCFD)	Q4 2026
Vertical playbooks (pharma, automotive, semiconductor)	2027
PRITHVEX v2.0 — autonomous, self-healing agent	2027

Conclusion

ENVIREX, powered by PRITHVEX, represents a shift from reactive supply chain firefighting toward intelligent, calibrated, and explainable risk anticipation. By pairing predictive analytics with provable uncertainty guarantees and transparent attribution, the platform gives organisations a defensible basis to act before a disruption escalates rather than after.

Presented by Team Envirex: Prannoy Sen, Piyush Rajan, Shaurya Kumar · github.com/PiyushRajan2007/ENVIREX-AI